

Analyzing β -glucuronidase expression in gut microbial populations of MS patients and healthy controls:

An intersection of immunology and computer science

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Abstract

Multiple sclerosis (MS) is a chronic and progressive autoimmune disorder of the central nervous system. Studies have observed specific differences in the microbiota of MS patients and some have established a clear link between estrogen variants and MS by observing a gender bias and demonstrating that estrogens have several immunoprotective and anti-inflammatory properties. β -glucuronidase is an exogenous enzyme that deconjugates estrogens into their active form and as such, may contribute to protection against MS. There has not yet been a study that characterizes the interplay between estrogen variants, β -glucuronidase, and the gut microbiome but doing so may help elucidate hormonal and immunological mechanisms underlying MS pathogenesis. We hypothesize that the relative gut microbial β -glucuronidase gene counts and by extension, gene expression, for healthy controls will be greater compared to that for MS patients. To test this hypothesis we assessed the estrobolome to analyze the levels of β -glucuronidase gene expression in control and MS patients. Using data obtained from previous clinical studies, we characterized microbiota populations through the QIIME2 pipeline and performed a predictive metagenomic analysis using PICRUSt2 plugin to generate β -glucuronidase gene counts and allow for pairwise comparison between MS patients and healthy controls. We were unable to confirm our hypothesis that gut microbial β -glucuronidase gene counts for healthy controls are greater than that of MS patients. We were however, able to qualitatively identify several microbial populations that could be correlated with MS pathogenesis which may prove worthwhile to further investigate.

Introduction

Multiple sclerosis (MS) is a chronic and progressive autoimmune disorder of the central nervous system (CNS). Host immune cells recognize and attack the oligodendrocytes, which produce the myelin sheaths necessary to propagate electrical impulses between neurons. T cells that are activated by myelin secrete cytokines, causing chronic inflammation and the blood-brain barrier to become penetrable to other immune cells. Repeated degradation of the myelin sheath leaves behind dysfunctional scar tissues and lesions called plaques that vary in their physiological symptoms depending on their location in the CNS.

Relapsing-remitting MS (RRMS) is by far the most prevalent type of MS, accounting for nearly 85 percent of all cases, and is characterized by bouts of autoimmune attacks that happen months or years apart.¹² There is currently no cure for MS; however, medications like corticosteroids, cyclophosphamide, IV immunoglobulin, and other immunosuppressants can be used to lessen the severity and frequency of relapses.¹ Several risk factors for MS include gender, Vitamin D deficiency, which has been shown to trigger peripheral T cells to cross the blood-brain barrier.^{2,3} The human gut microbiota also associated with immune disorders like MS because dysbiosis has been

shown to disrupt its normally immunoprotective mechanisms. Consisting of over 100 trillion microorganisms, the gut microbiota harbors many microorganisms that play crucial roles in a large range of physiological functions like digestion and immune activity.

Studies have observed that there are specific microbial taxonomic differences in the gut microbiome of MS patients when compared to healthy controls.⁴ Some are associated with the promotion of inflammatory cytokines and overall inflammation. In other studies with experimental allergic encephalomyelitis (EAE), the most widely used animal model to study MS, it has been reported that T regulatory (Treg) cells become increasingly dysfunctional as a result of broad-spectrum antibiotic intervention.⁴ This supports the hypothesis that the gut microbial composition may play a large role in MS pathogenesis.

The factor driving the gender bias characteristic of RRMS and its onset overall may be attributed to the influence of gut microbes and gender-based autoimmunity. In pathogen-free female mice, the gender bias of MS pathogenesis can be reversed by transferring gut microbiota from male mice, suggesting an inherent gender-based difference in gut microbiome composition.⁵ Moreover, there is clinical evidence that identifies a clear link between estrogen variants and MS.⁵ 17 β -estradiol (E2) and estriol (E3) are endogenous estrogens produced in humans. E2 inhibits pro-inflammatory cytokines and E3 has been shown to reduce EAE severity in mouse models.⁵ A randomized, placebo controlled, double-blind trial conducted with female RRMS patients and E3 found that confirmed relapse rate was lower in the E3-treated RRMS group.⁵ As such, lower concentrations of active circulating estrogens may promote a pro-inflammatory breeding ground for MS pathogenesis.

Here, we focus on a subset of the microbiome called the estrobolome which concerns bacteria that are capable of modulating or metabolizing the body's circulating estrogens. Bacteria in the estrobolome regulate circulating estrogen levels by secreting β -glucuronidase, an enzyme that deconjugates estrogen into their active form.⁶ Subsequent dysbiosis of the gut microbiome reduces deconjugation and decreases circulating estrogens. This may contribute to the development of MS because estrogen variants, as previously mentioned, have been shown to have several immunoprotective and anti-inflammatory properties.

There has not yet been a study that characterizes the interplay between estrogen variants and the gut microbiome but doing so may help elucidate hormonal and immunological mechanisms underlying MS pathogenesis. To that point, in this study, we assess and characterize the estrobolome in MS and control patients to identify the potential role of β -glucuronidase gene expression on the pathogenesis of MS.

Our hypothesis is that the relative gut microbial β -glucuronidase gene counts and by extension, gene expression, for healthy controls will be greater compared to that for MS patients.

Specific Aims

Our overarching aim is to assess and characterize the estrobolome in RRMS and control patients to identify the impact of β -glucuronidase gene expression on the pathogenesis of MS. We plan to specifically to:

- **Aim 1:** Identify what microbial populations are associated with RRMS.
- **Aim 2:** Identify what microbial populations in the estrobolome express the β -glucuronidase gene by assessing DNA counts in case vs. control.

Methods

Clinical Studies

When choosing studies for viable datasets, we analyzed their study design through their exclusion criteria and methods. The first clinical study recruited 60 MS patients from the Partners MS Center at Brigham and Women’s Hospital and 43 controls from the Brigham and Women’s Hospital PhenoGenetic project.⁷ Their exclusion criteria is comprehensive and as follows: no antibiotic use in the prior six months, no probiotic or corticosteroid use, no history of using immunosuppressive medications, no history of gastroenteritis or autoimmune disease, and no travel outside of the country in the prior month. None of the MS patients had an active relapse at the time of sampling, and pregnant women were excluded. Some elements of the sample population to note include that the control population consisted of proportionally more females than males (86 percent to 14 percent) whereas the gender disparity for the MS population, although still apparent, was significantly reduced (68 percent to 32 percent). All subjects were Caucasian except for two Black and one Hispanic MS patients. For sample collection, stool samples were collected from the subjects in stool collection containers at any time of day without dietary restrictions. DNA was extracted from the stool samples using the PowerSoil DNA Isolation Kit. To assess only bacterial DNA, they amplified bacterial DNA with a specific probe for 16s rRNA. Finally, in order to assess the microbial community structure of the samples, they generated DNA libraries for 16s rRNA using the V3-V5 regions. This enabled them to later amplify and sequence the V4 regions with Next Generation Sequencing and thus identify which microbial populations were present in the samples.

The second clinical study recruited 45 patients with MS, none of whom had an active relapse at the time of study, and 44 healthy controls.⁹ Their exclusion criteria is as follows: no patients suffering from infectious diseases and no antibiotic treatment during the collection of fecal samples. From these patients, they obtained 16s rRNA and metagenomic sequences from fecal samples which were then transported at 4 degrees C to the laboratory in a plastic bag containing a disposable oxygen-absorbing and carbon dioxide-generating agent so anaerobes sensitive to oxygen would survive the trip. Once in the lab, the fecal samples were suspended in phosphate-buffered saline containing 20 percent glycerol, immediately frozen using liquid nitrogen and stored at -80 degrees C until use. The bacterial DNA was isolated and purified using enzymatic lysis methods, presumably similar to the PowerSoil DNA Isolation Kit used in the first clinical study. They sequenced the V1-V2 regions of the 16s rRNA gene and amplified them by PCR. They then used whole genome sequencing analysis with IonProton platforms to obtain metagenomic sequences.

Table 1. Comparison of the Methods of the Clinical Studies

Title of publication	Alterations of the human gut microbiome in multiple sclerosis (Jangi et al) ⁷	Gut microbiome of treatment-naïve MS patients of different ethnicities early in disease course (Ventura et al) ⁹
Experimental Design and Exclusion Criteria	Untreated patients were treatment naïve or received no steroid treatment in the previous month. Subjects were to have had no other treatments in the past six months including antibiotic or probiotic use and corticosteroids. They were also required to have no history of gastroenteritis, irritable bowel syndrome, bowel surgery, inflammatory bowel disease or other autoimmune diseases where immunosuppressive medications were taken. Subjects could not be pregnant nor could they have travelled to outside countries in prior months. None of the MS patients had active relapse, and treated patients received beta-interferon or glatiramer acetate for at least six months leading up to the study.	Subjects of mixed races were not included and all subjects were between the ages of 18 and 70. The only allowed medications were NSAIDs and analgesics and all subjects were to have had no prior or current use of disease modifying or biologic therapies for BRX, recent use of antibiotic therapy, extreme diet, inflammatory bowel disease, or prior GI tract surgery which all may have left permanent residues on their microbiome.
Number of Subjects	60 MS patients (57 Caucasian, 1 Hispanic and 2 African American) and 43 healthy controls. In the context of our study, we only analyzed the untreated MS patients (n=27), excluding patients undergoing beta-interferon or glatiramer acetate treatment.	45 MS subjects (15 Caucasian, 16 Hispanic and 14 African American) and 44 healthy controls
Sequencing Method	16s Sequencing	Whole Genome Sequencing
Sample Type	Stool	Stool
Notable Points	MS cohort had more males	Three stool samples from African Americans were collected more than six months after diagnosis.
	Most of the subjects were Caucasian; there were only two Black MS patients and one Hispanic MS patient	

Data Analysis

We used the QIIME2 pipeline to identify microbial populations present in the datasets retrieved from the two clinical studies, which contained raw 16s rRNA sequences. We imported the sample data and screened them using DADA2 for demultiplexing and denoising before utilizing the pipeline to align the resulting sequences with the GreenGene database. This enabled us to quantify core metrics like alpha and beta diversity while also developing a phylogenetic tree to identify observed OTUs as well as their differential abundance. For alpha diversity, we considered the Faith, Shannon

and Observed Features metrics to a sequencing depth of 50,000 base pairs. For beta diversity, the UniFrac method we used is a good fit because it measures the phylogenetic distances between sets of taxa based on mutual lineages while taking into account degrees of similarity between 16s rRNA sequences.

With the alignment data, we then used the PICRUST plugin in QIIME2 to perform a predictive analysis on the raw 16s rRNA sequences. Both of these analyses generate β -glucuronidase gene counts specific to each microbial population present for pairwise comparison between MS patients and healthy controls.

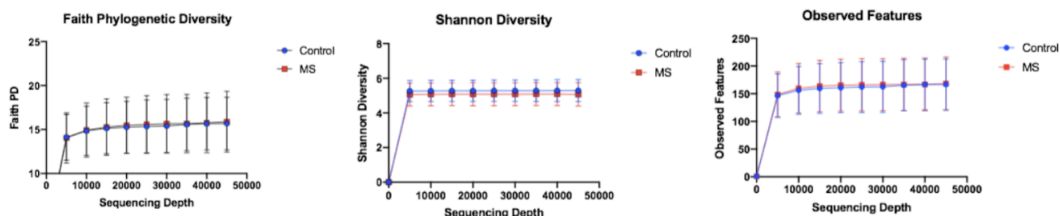
Results

We visualized our results using QIIME2 View, but we re-visualized several figures using Graph-Pad Prism.

Across all three alpha diversity metrics, there was no significant difference in alpha diversity between MS patients and healthy controls regardless of the sequencing depth. This trend was also observed in the clinical study we retrieved the data from. This suggests that similar microbial populations are present in the gut microbiomes of MS patients and healthy controls.

To assess differences in community structure (beta diversity) we used the emperor PCoA plots generated with QIIME2 to identify the differential microbial diversity between MS patients and controls. Our PCoA for weighted data has two clusters of healthy controls and MS patients (Figure 1b). Weighted data takes into account phylogenetic diversity by mapping the sequences onto a tree for comparison whereas unweighted data looks at the sequences at face value. We then performed unpaired, nonparametric Kolmogorov-Smirnov tests on the data, obtaining an insignificant p-value for unweighted data but a p-value of 0.003 for weighted data.

Alpha and beta diversity analyses are helpful because alpha diversity is inversely related to dysbiosis and MS pathogenesis while beta diversity is helpful to distinguish between separate microbial communities and characterize differences in the microbiota of MS and control patients.



Next, we assessed the microbial composition of each subject to identify taxonomic groups present in each subject and their differential abundance. Each column represents a subject, and each color represents a specific OTU. The wider each column is, the more abundant the taxa is.

We then identified OTUs that were more abundant in one group than the other for further analysis regarding how they may have an effect on MS pathogenesis (Figure 2). *Akkermansia muciniphila* and *Bifidobacterium adolescentis* were abundant in untreated MS cases while the other OTUs were abundant in healthy controls.

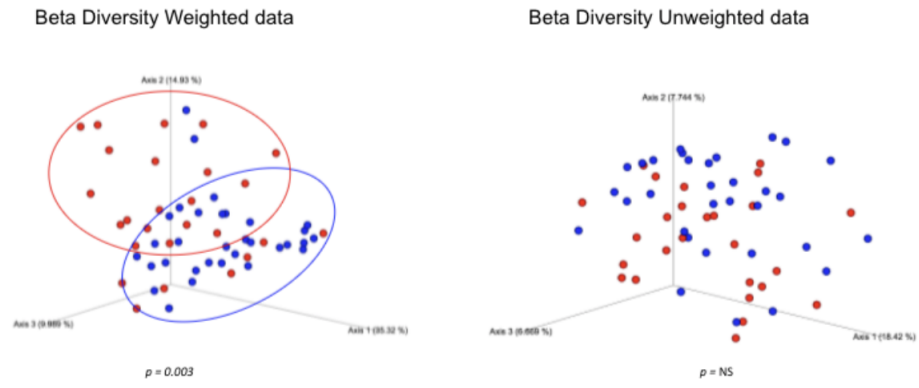
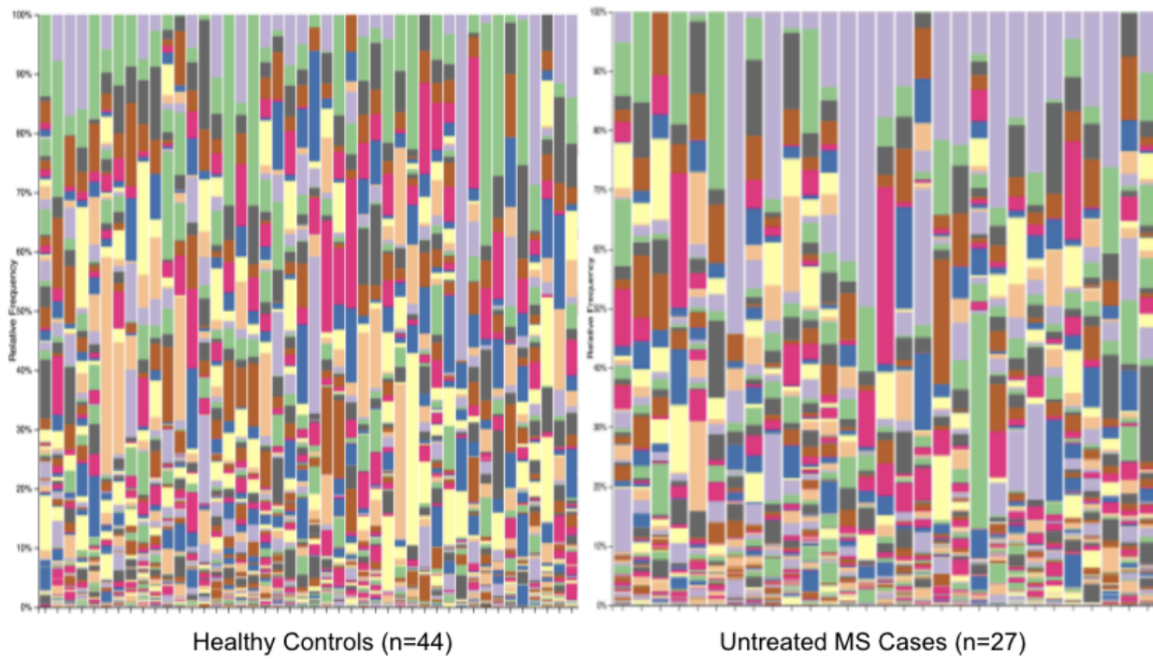


Figure 1: a) Comparison of alpha diversity metrics between MS and control patients. b) Difference in phylogenetic distances between MS patients and controls regarding community structure.



Legend

- (1) *g_Akkermansia; s_muciniphila*
- (1) *g_Bacteroides*
- (2) *g_Dialister*
- (3) *g_Bifobacterium; s_adolescentis*
- (3) *g_Prevotella; s_copri*
- (6) *g_Bacteroides; s_eggerthii*
- (7) *g_Bacteroides; s_plebeius*

Figure 2: Microbial species present and their differential abundance in healthy controls and untreated MS cases

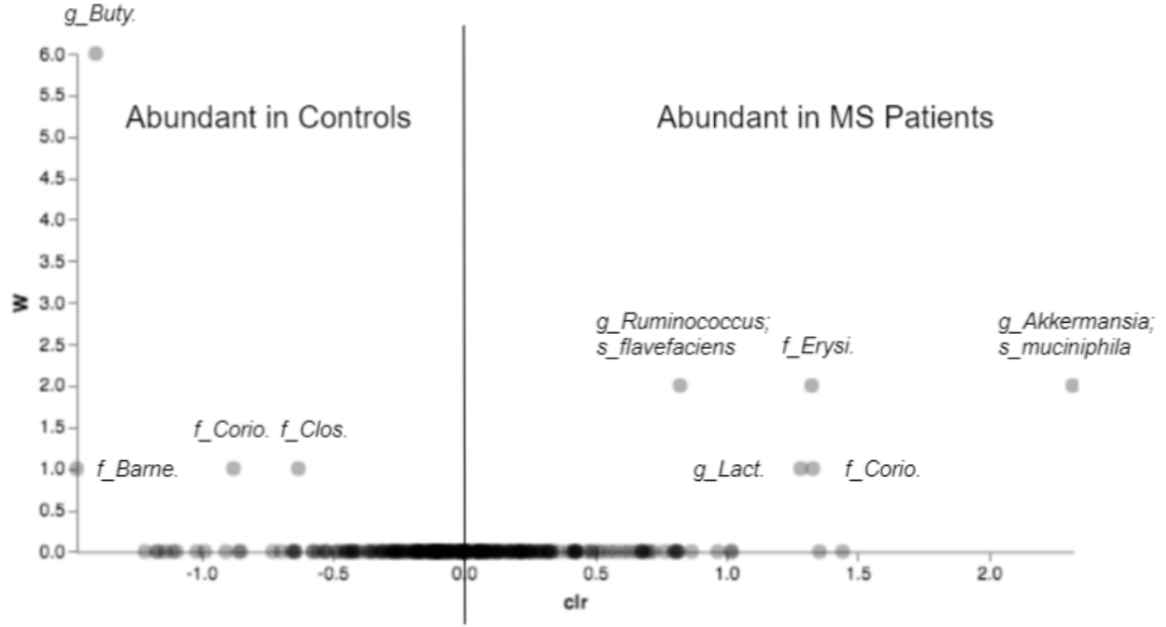


Figure 3: L7 Volcano Plot of the differential abundance of OTUs present in MS and control subjects

Additionally, we quantitatively assessed the differential abundance of OTUs present in the 71 subjects, identifying specific OTUs that were more abundant in either MS patients or healthy controls. The y axis (W) refers to absolute abundance. The centered log ratio (clr) normalizes high composition data sets and in our study, a positive ratio indicates the associated OTU is significantly more abundant in MS patients whereas a negative clr indicates the opposite, that the OTU is significantly more abundant in healthy controls. Notable OTUs abundant in healthy controls include microorganisms in the family Butyricimonas and Barnesiellaceae and notable OTUs abundant in MS patients include Akkermansia muciniphila and *Ruminococcus flavefaciens*.

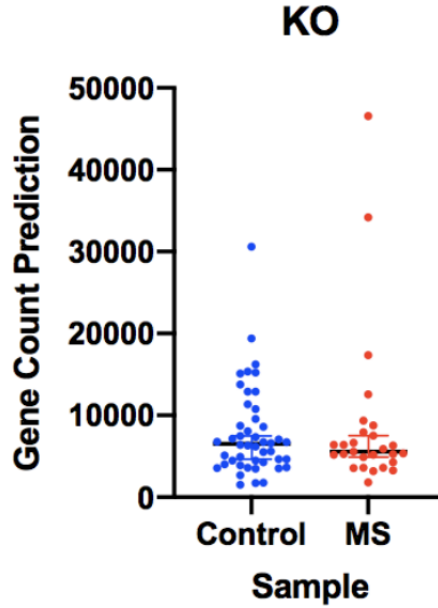


Figure 4: β -glucuronidase gene count metagenome prediction comparison between control and MS patients

Finally, we assessed β -glucuronidase gene count through PICRUST2 metagenome prediction analysis and performed an unpaired, nonparametric Kolmogorov-Smirnov test on our gene counts and obtained an insignificant p-value. However, it does appear that the median gene count for controls is higher than that for MS patients.

It is important to note too that these predictive analyses based on differential abundance testing can vary between amplicon-based metagenome predictions like in our case and those utilizing shotgun metagenomics sequencing data.

Discussion

In this study, we explored the estrobolome to characterize the estrobolome in RRMS and control patients and the impact of the expression of the β -glucuronidase gene on MS pathogenesis. We identified microbial populations associated with RRMS and those that express the β -glucuronidase gene by performing a pairwise analysis of β -glucuronidase gene count between healthy controls and MS patients. This was done using datasets from two previous clinical studies and by taking advantage of bioinformatics platforms like QIIME2 to assess differences in microbial communities and structure and other state-of-the-art pipelines like PICRUST for predictive metagenomic analysis of the β -glucuronidase gene. Because β -glucuronidase deconjugates estrogens, which have been shown to have anti-inflammatory and immunoprotective properties, we hypothesized that the relative gut microbial β -glucuronidase gene counts and by extension, gene expression, for healthy controls will be greater compared to that for MS patients.

We first assessed the alpha diversity of all of our samples obtained from the two clinical studies, one dataset generated from 16s sequencing and the other from whole genome sequencing, using the Faith, Shannon, and Observed Features metrics up to a sequencing depth of 50,000 and were able to confirm observations made by other studies that similar microbial populations are present in the gut microbiomes of MS patients and healthy controls. There was a significant phylogenetic difference in microbial community structure between the microbiome of MS patients compared with

that of healthy controls (Figure 1b), $p < 0.05$.

Since there were differences in microbial community structure, we then assessed the microbial composition of each subject to identify specific taxonomic groups present in each subject and their differential abundance. With this and a further analysis on the differential abundance of these OTUs, we were able to qualitatively and quantitatively identify several microbial populations that were more abundant in either MS patients or healthy controls.

Additionally, we were not able to confirm our hypothesis that gut microbial β -glucuronidase gene counts for healthy controls are greater compared to that for RRMS patients because the p-value for the t-tests for our gene counts between the two groups was not significant. This suggests that β -glucuronidase gene expression may not be as directly correlated to MS pathogenesis as we had predicted.

A caveat of our analyses, especially our gene count predictive analysis, is that we were limited by the computing power of our personal computers and bottlenecked by downloading large files for other analyses. Because of this computational capacity and limited time, we were unable to perform whole genome analysis of the alignment data calibrated to population abundance using MetaPhlAn and HUMAnN on the confirmed microbial populations. A potential future direction for this study would be to secure increased computing power in order to consider more subjects during predictive analyses, analyze the whole genome data sets, and further analyze the impact of selected OTUs. It may prove worthwhile to study these microbial populations in greater detail to better understand their physiological significance in the human body, particularly in the estrobolome, as they may be correlated with MS pathogenesis or confer protection against MS.

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